

6.1 Linear Equations**6.1.1 Recall Linear Equation in one Variable:**

If symbol of equality “=” is involved in an open sentence then such sentence is called an **equation**. Linear equations with one variable i.e. $ax+b=0$, $a \neq 0$, are equations where variable has an exponent “1” which is typically not shown.

6.1.2 Solve linear equations with Rational Coefficients:

The value of the unknown (variable) for which the given equation becomes true is called a solution or root of the equation.

Example 01 Solve: $3x-1=5$

Solution: $3x-1=5$

$$\Rightarrow 3x = 5+1$$

$$\Rightarrow x = \frac{6}{3}$$

$$\Rightarrow x = 2$$

Thus, the solution set is $\{2\}$

Example 02 Solve: $\frac{2}{3}(x+3) = 3 + \frac{5x}{9}$

Solution: $\frac{2}{3}(x+3) = 3 + \frac{5x}{9}$

$$9 \times \frac{2}{3}(x+3) = 9 \times 3 + 9 \times \frac{5x}{9} \quad (\text{Multiplying both sides by 9})$$

$$\Rightarrow 3 \times 2(x+3) = 27 + 5x$$

$$\Rightarrow 6(x+3) = 27 + 5x$$

$$\Rightarrow 6x + 18 = 27 + 5x$$

$$\Rightarrow 6x - 5x = 27 - 18$$

$$\Rightarrow x = 9$$

Thus, the solution set is $\{9\}$.

Example 03 Age of father is 13 times the age of his son. It will be only five times after four years. Find their present ages.

Solution: Let present age of son = x years,
and present age of father = $13x$ years,

According to given condition,

$$\therefore 13x + 4 = 5(x + 4)$$



$$\begin{aligned}
 &\Rightarrow 13x + 4 = 5x + 20 \\
 &\Rightarrow 13x - 5x + 4 = 5x - 5x + 20 \quad (\text{Subtracting } 5x \text{ from both sides}) \\
 &\Rightarrow 8x + 4 = 20 \\
 &\Rightarrow 8x + 4 - 4 = 20 - 4 \quad (\text{Subtracting } 4 \text{ from both sides}) \\
 &\Rightarrow 8x = 16 \\
 &\Rightarrow \frac{8x}{8} = \frac{16}{8} \quad (\text{Dividing } 8 \text{ on both sides}) \\
 &\Rightarrow x = 2
 \end{aligned}$$

Hence present age of father = $13 \times 2 = 26$ years
and present age of son = 2 years.

Example 04 When 16 is added to $\frac{1}{3}$ of number the result is $2\frac{1}{3}$ of the original number. Find the number?

Solution: Let x be the number, the according to the given condition:

$$\begin{aligned}
 &16 + \frac{1}{3}x = 2\frac{1}{3}x \\
 &\Rightarrow 16 + \frac{1}{3}x = \frac{7}{3}x \\
 &\Rightarrow 16 = \frac{7}{3}x - \frac{1}{3}x \\
 &\Rightarrow 16 = \left(\frac{7}{3} - \frac{1}{3}\right)x \\
 &\Rightarrow 16 = \left(\frac{7-1}{3}\right)x \\
 &\Rightarrow 16 = \frac{6}{3}x \\
 &\Rightarrow 16 \times 3 = 6x \\
 &\Rightarrow \frac{48}{6} = x \quad \Rightarrow x = 8
 \end{aligned}$$

6.1.3 Reduce Equations involving radicals to Simple linear Form and find their solutions.

Definition: An equation in which the variable appears under the radical sign, is called the radical equation.

For example, $3\sqrt{t} - \sqrt{t+1} = 2$ and $\sqrt{x} = 8$ are radical equations.



Solution of radical equation is explained with help of the following example.

Example 01 Solve: $\sqrt{2x+11} = \sqrt{3x+7}$

Solution: $\sqrt{2x+11} = \sqrt{3x+7}$

Squaring on both the sides, we have,

$$(\sqrt{2x+11})^2 = (\sqrt{3x+7})^2$$

$$\Rightarrow 2x+11 = 3x+7$$

$$\Rightarrow 2x-3x = 7-11$$

$$\Rightarrow -x = -4$$

$$\Rightarrow x = 4$$

Verification: Put $x = 4$ in the given equation, then,

$$\sqrt{2(4)+11} = \sqrt{3(4)+7}$$

$$\text{or } \sqrt{8+11} = \sqrt{12+7}$$

$$\text{or } \sqrt{19} = \sqrt{19}$$

Thus, solution set is $\{4\}$.

- Notes:** 1. Sometimes the obtained root from radical equation does not satisfy the original equation, it is called an extraneous root.
2. Solutions of radical equations must be verified.

Exercise 6.1

1. Solve the following equations

(i) $\frac{1}{4}x = 5$

(ii) $\frac{x}{4} = -3$

(iii) $-5 = \frac{-x}{6}$

(iv) $\frac{-x}{8} = -5$

(v) $y - \frac{2}{5} = -\frac{1}{3}$

(vi) $2y - \frac{3}{5} = \frac{1}{2}$

(vii) $\frac{2x-4}{5} = \frac{5x-12}{4}$

(viii) $\frac{3x}{5} + 7 = \frac{2x}{3}$

(ix) $\frac{3x}{5} + 7 = \frac{2x}{3} + \frac{4x}{5}$

(x) $\frac{6}{2x-5} - \frac{4}{x-3} = 0$

(xi) $\frac{7x-4}{15} = \frac{7x+4}{10}$

(xii) $\frac{3x-2}{10} = \frac{7x-3}{15} - 2$

(xiii) $\frac{12x-3}{12} = \frac{12x+3}{8}$

(xiv) $\frac{1}{4}x + x = -3 + \frac{1}{2}x$

(xv) $\frac{1}{3} + 2m = m - \frac{3}{2}$



2. When 25 added to a number, the result is halved; the answer is 3 times the original number. What is the number?
3. When a number is added to 4, the result is equal to subtracting 10 from 3 times of it. What is the number?
4. Bilal is 6 year older than Ali, Five years from now the sum of their age will be 40. How old are both of them.
5. **Find the Solution set of the following equations and also verify the answer:**

(i) $6 + \sqrt{x} = 7$

(ii) $\sqrt{x-9} = 1$

(iii) $\sqrt{\frac{y}{4}} - 2 = 3$

(iv) $\sqrt{4x+5} = \sqrt{3x-7}$

(v) $\frac{\sqrt{3y+12}}{7} = 3$

(vi) $\sqrt{x} + 9 = 7$

(vii) $\sqrt{25y-50} = \sqrt{y-2}$

(viii) $\sqrt{x} - 8 = 1$

(ix) $10\sqrt{x+20} = 100$

6.2 Equations involving absolute values

6.2.1 Define absolute values

The absolute value of a real number x is denoted by $|x|$, is the distance of x from zero, either from left or from right of zero.

If x is real number, then absolute value or modulus value of x is denoted by $|x|$, is defined as under:

$$|x| = \begin{cases} x, & \text{when } x > 0 \\ 0, & \text{when } x = 0 \\ -x, & \text{when } x < 0 \end{cases}$$

Example $|-5| = 5, | +7 | = 7, \left| -\frac{1}{2} \right| = \frac{1}{2}, |0| = 0$ and so on.

Note: The absolute value of a number is always non negative.



6.2.2. Solve the Equations, involving absolute values in one variable

Example 01 Find the solution set of $|5x-3|-2=3$ **Solution:** Given that

$$|5x-3|-2=3$$

$$\Rightarrow |5x-3|=5$$

By the definition of modulus, we have,

$$5x-3=5 \quad \text{or} \quad 5x-3=-5$$

$$\Rightarrow 5x=5+3 \quad \text{or} \quad 5x=-5+3$$

$$\Rightarrow 5x=8 \quad \text{or} \quad 5x=-2$$

$$\Rightarrow x=\frac{8}{5} \quad \text{or} \quad x=-\frac{2}{5}$$

Thus, the solution set is $\left\{\frac{8}{5}, -\frac{2}{5}\right\}$ **Example 02** Find the solution set of $|5x-3|+7=3$ **Solution:** Given that

$$|5x-3|+7=3$$

$$\Rightarrow |5x-3|=-4$$

The modulus of a real number never be negative

 $\therefore$ Solution set = $\{ \}$ **Example 03** Solve $|5x-3|-2=3$, where $x \in W$.**Solution:** Given that,

$$|5x-3|-2=3$$

$$\Rightarrow |5x-3|=5$$

By the definition of modulus, we have,

$$\text{so, } 5x-3=5 \quad \text{or} \quad 5x-3=-5$$

$$\Rightarrow 5x=5+3 \quad \text{or} \quad 5x=-5+3$$

$$\Rightarrow 5x=8 \quad \text{or} \quad 5x=-2$$

$$\Rightarrow x=\frac{8}{5} \quad \text{or} \quad x=-\frac{2}{5}$$

$$-\frac{2}{5} \text{ and } \frac{8}{5} \notin W$$

Thus, the solution set is $\{ \}$



Example 04 Solve $|2y - 5| + 2 = 7$

Solution: Given that $|2y - 5| + 2 = 7$

$$\Rightarrow |2y - 5| = 7 - 2$$

$$\Rightarrow |2y - 5| = 5$$

By definition of modulus we have,

$$\text{so, } 2y - 5 = 5 \quad \text{or} \quad 2y - 5 = -5$$

$$\Rightarrow 2y = 5 + 5 \quad \text{or} \quad 2y = -5 + 5$$

$$\Rightarrow 2y = 10 \quad \text{or} \quad 2y = 0$$

$$\Rightarrow y = \frac{10}{2} \quad \text{or} \quad y = \frac{0}{2}$$

$$\Rightarrow y = 5 \quad \text{or} \quad y = 0$$

Thus, the solution set is $\{5, 0\}$.

Exercise 6.2

Find the solution set of the following equations.

1. $|2x + 1| = 6$

2. $|5x - 12| = 7$, where $x \in W$

3. $\left|\frac{2x}{7}\right| = 12$

4. $\left|\frac{2x+1}{3}\right| = 8$

5. $|5x - 3| - 8 = 4$, where $x \in N$

6. $\left|\frac{5x+1}{7}\right| - 3 = 8$

7. $\left|\frac{2x+3}{4}\right| + 2 = 7$

8. $\left|\frac{3x+6}{12}\right| + 1 = 3$, where $x \in Z$

9. $\frac{3}{2} = |7x + 8|$

10. $\left|\frac{2x-3}{5}\right| - 12 = 5$

11. $|3x + 1| + 1 = \frac{3}{4}$

12. $\left|\frac{2x+1}{7}\right| = 1$

6.3 Linear inequalities

A linear algebraic expression which contains the sign of inequality is called linear inequality or linear inequation.

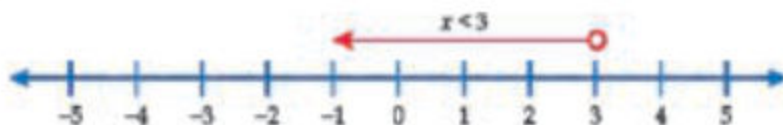
6.3.1 Define inequalities ($>$, $<$) and ($\geq$, $\leq$).

The following relational operators are called inequalities.

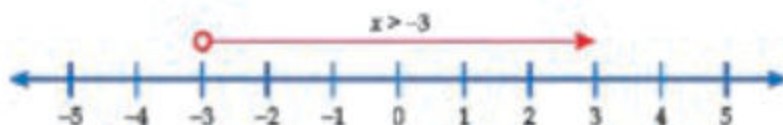
- ' $<$ ' means less than,
- ' $>$ ' means greater than,
- ' $\leq$ ' means less than or equal to,
- ' $\geq$ ' means greater than or equal to.



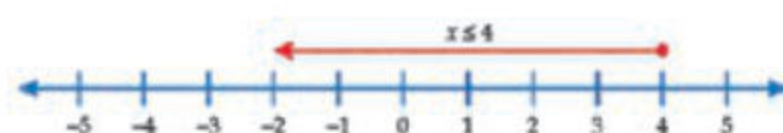
Example 01 Illustrate $x < 3$ on the number line



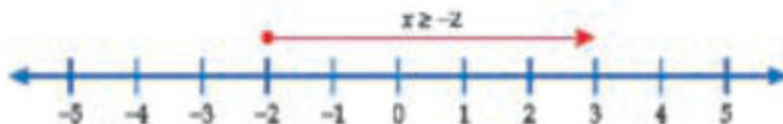
Example 02 Illustrate $x > -3$ on the number line



Example 03 Illustrate $x \leq 4$ on the number line



Example 04 Illustrate $x \geq -2$ on the number line



Note: Hollow circle 'O' shows that number is not included and dark circle '●' shows that number is included.

6.3.2 Recognize properties of inequalities (trichotomy, transitive, additive, multiplicative).

The following are some important properties of inequalities.

(i) **Trichotomy Property:**

For any two real numbers a and b , one and only one statement of the following is always true.

$$a < b, a = b \text{ or } a > b$$

(ii) **Transitive Property:**

For any three real numbers a, b and c

$$\text{If } a < b \text{ and } b < c \Rightarrow a < c$$

$$\text{and } a > b \text{ and } b > c \Rightarrow a > c$$



(iii) **Additive Property:**

For any three real numbers

if $a > b$ then $a + c > b + c$, $\forall a, b, c \in \mathbb{R}$

or if $a < b$ then $a + c < b + c$, $\forall a, b, c \in \mathbb{R}$

(iv) **Multiplicative Property:**

(a) If $a > b$ then $ac > bc$, $\forall a, b, c \in \mathbb{R}$ and $c > 0$

or If $a < b$ then $ac < bc$, $\forall a, b, c \in \mathbb{R}$ and $c > 0$

(b) If $a > b$ then $ac < bc$, $\forall a, b, c \in \mathbb{R}$ and $c < 0$

or If $a < b$ then $ac > bc$, $\forall a, b, c \in \mathbb{R}$ and $c < 0$

6.4 Solving linear Inequalities

6.4.1 Solving linear inequalities with rational coefficients.

The following examples will help us understand the solution and show on the number line.

Example 01 Find the solution set of $3x + 1 < 7$ $\forall x \in W$, and show on the number line

Solution:

Given that

$$3x + 1 < 7 \quad \forall x \in W$$

$$(3x + 1) - 1 < 7 - 1$$

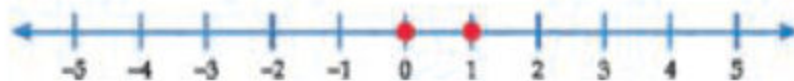
$$3x < 6$$

$$x < \frac{6}{3}$$

$$x < 2$$

Therefore, the solution set is $\{x | x \in W \wedge x < 2\} = \{0, 1\}$

The solution is illustrated on the number line as under:



Example 02 Find the solution set of $x - 11 \leq 9 - 4x \quad \forall x \in \mathbb{Z}$ and show on the number line

Solution: Given that $x - 11 \leq 9 - 4x \quad \forall x \in \mathbb{Z}$

$$x - 11 \leq 9 - 4x$$

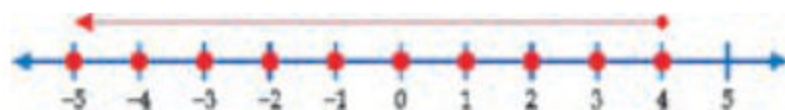
$$x + 4x \leq 9 + 11$$

$$5x \leq 20$$

$$x \leq \frac{20}{5}$$

$$x \leq 4$$

Therefore, the solution set is $\{x \mid x \in \mathbb{Z} \wedge x \leq 4\} = \{\dots, -2, -1, 0, 1, 2, 3, 4\}$



Example 03 Find the solution set of $2x + 5 > 7 \quad \forall x \in \mathbb{R}$. Also illustrate the solution on the number line.

Solution: Given that

$$2x + 5 > 7 \quad x \in \mathbb{R}$$

$$\text{or } 2x > 7 - 5$$

$$\text{or } 2x > 2$$

$$\text{or } x > 1$$

Thus, the solution set is $\{x \mid x \in \mathbb{R} \wedge x > 1\}$

The solution on number line is illustrated as under:



Example 03 Find the solution set of $-6 < 2x + 1 < 11, \quad \forall x \in \mathbb{Z}$. Also express it on number line.

Solution: Given that $-6 < 2x + 1 < 11, \quad \forall x \in \mathbb{Z}$.

Splitting the inequality as under:

$$-6 < (2x + 1) \quad \text{and} \quad 2x + 1 < 11$$

$$\text{or } -6 - 1 < 2x \quad \text{and} \quad 2x + 1 < 11 - 1$$

$$\text{or } -7 < 2x \quad \text{and} \quad 2x < 10$$



$$\text{or } \frac{-7}{2} < x \quad \text{and} \quad x < 5$$

Thus, the solution set is $\{x | x \in \mathbb{Z} \wedge -\frac{7}{2} < x < 5\} = \{-3, -2, -1, 0, 1, 2, 3, 4\}$

The solution on number line is illustrated as under:



Example 04 Ayesha scored 78, 72 and 86 on the first three out of four tests. What score must be recorded on the fourth test to have average at least of 80?

Solution: let score of the fourth test be x so that.

$$\frac{78 + 72 + 86 + x}{4} \geq 80$$

$$78 + 72 + 86 + x \geq 320$$

$$236 + x \geq 320$$

$$x \geq 320 - 236$$

$$x \geq 84$$

Ayesha must score 84 on the fourth test to maintain average of 80.

Exercise 6.3

1. Find the solution sets of the following inequalities and also illustrate the solution on the number line.

(i) $2x - 7 > 6 + x \quad \forall x \in \mathbb{N}$

(ii) $7x - 6 > 3x + 10, \quad \forall x \in \mathbb{R}$

(iii) $\frac{y + 5}{20} < \frac{25 - 4y}{10}, \quad \forall y \in \mathbb{N}$

(iv) $|2x + 3| < x + 2, \quad \forall x \in \mathbb{Z}$

(v) $|2y + 8| < 11, \quad \forall y \in \mathbb{R}$

(vi) $5(2y - 3) > 6(y - 8), \quad \forall y \in \mathbb{R}$

2. Ali scored 66 and 72 marks respectively. For his two Tests, what is the lowest mark he must have scored for his third test If an average score of at least 75 is required to qualify for a bonus prize

3. Seven less than three times the sum of a number and 5 is at least 10, Find all the number that satisfy this condition.

Review Exercise 6

1. True and false questions

Read the following sentences carefully and encircle 'T' in case of true and 'F' in case of false statement.

- (i) $ay + b = 0$, where $a = 0$ is a linear equation T/F
- (ii) The solution set of $3y - 2 < 7$, $y \in \mathbb{N}$ is $\{4, 5, 6, \dots\}$ T/F
- (iii) The solution set of $\sqrt{y} + 1 = 3$ is $\{4\}$. T/F
- (iv) The solution set of $|4y| = 8$ is $\{2, -2\}$. T/F
- (v) The solution set of $-2 \leq x \leq 2$, $x \in \mathbb{Z}$ is $\{-2, 0, 2\}$. T/F

2. Fill in the blanks.

- (i) The solution set of $2y = -y$ is _____.
- (ii) The solution set of $\sqrt{y+5} = 5$ is _____.
- (iii) The solution set of $|x| - 4 = 0$ is _____.
- (iv) The solution set of $\sqrt{x+5} + 2 = 4$ is _____.
- (v) The solution set of $0 < y + 2 < 5$ when $y \in \mathbb{R}$ is _____.

3. Tick (✓) the correct answer

- (i) The solution set of linear equation in one variable has
 - (a) One solution
 - (b) Two solutions
 - (c) Three solutions
 - (d) More than one solutions
- (ii) $|-20|$
 - (a) $= 20$
 - (b) < 20
 - (c) $= -20$
 - (d) > 20
- (iii) $x \leq 4$ means
 - (a) $x < 4$
 - (b) $x = 4$
 - (c) $x < 4$ or $x = 4$
 - (d) $x > 4$ or $x = 4$
- (iv) The solution set of $\sqrt{y} = 10$ is
 - (a) $\{100\}$
 - (b) $\{10\}$
 - (c) $\{-10\}$
 - (d) $\{-10, 10\}$
- (v) $\sqrt{y+4} + 2 = 8$ is a
 - (a) Linear equation
 - (b) Radical equation
 - (c) Cubic equation
 - (d) Quadratic equation



- (vi) The solution set of $5-3y = -7$ is
 (a) $\{-4\}$ (b) $\{1, 4\}$
 (c) $\{4\}$ (d) $\{12\}$
- (vii) The solution set of $\sqrt{5y+5}+5 = 10$ is
 (a) $\{\pm 4\}$ (b) $\{5\}$
 (c) $\{4\}$ (d) $\{-4\}$
- (viii) The solution set of $\left|\frac{5y}{3}\right| = 5$ is
 (a) $\{3\}$ (b) $\{-5, 5\}$
 (c) $\{3, -3\}$ (d) $\{-3\}$
- (ix) The solution set of $|-y| = 0$ is
 (a) $\{1\}$ (b) $\{-1\}$
 (c) $\{0\}$ (d) $\{\}$
- (x) If $x > 0$, $y > 0$ and $x-y < 0$, then which of the following relation holds good?
 (a) $x < y$ (b) $x + y < 0$
 (c) $x > y$ (d) $y - x < 0$





Summary

- ◆ An equation of the form $ax + b = 0$, where $a, b \in \mathbb{R}$ and $a \neq 0$ is called a linear equation.
- ◆ An equation in which the variable appears under the radical sign, is called a radical equation. Radical equation can have extraneous roots, hence verification of the solution is essential.
- ◆ If $x \in \mathbb{R}$ then, $|x| = \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ -x & x < 0 \end{cases}$
- ◆ If $x, y \in \mathbb{R}$ then
 - (i) $|x| \geq 0$ (ii) $|-x| = |x|$ (iii) $|xy| = |x| \cdot |y|$
 - (iv) $\left| \frac{x}{y} \right| = \frac{|x|}{|y|}$ (v) $|x| = b$ then $x = b$ or $x = -b$
- ◆ For inequality, we use $<, >, \leq, \geq$.
- ◆ A linear algebraic expression which contains the sign of inequality is called linear inequality or inequation.
- ◆ Properties of inequalities:
 - (i) $a < b$ or $a = b$ or $a > b, \forall a, b \in \mathbb{R}$ (Trichotomy)
 - (ii) $a > b$ and $b > c \Rightarrow a > c, \forall a, b, c \in \mathbb{R}$ (Transitive)
 - (i) $a > b, c > 0 \Rightarrow ac > bc$ and $\frac{a}{c} > \frac{b}{c}, \forall a, b, c \in \mathbb{R}$ (Multiplication and Division Properties)

Unit

7

• Weightage = 5%

LINEAR GRAPH AND THEIR APPLICATION

Student Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- ◆ Identify pair of real numbers as an ordered pair.
- ◆ Recognize an ordered pair through different examples; for instance, an ordered pair (2, 3) to represent a seat, located in an examination hall, at the intersection of 2nd row and 3rd column.
- ◆ Describe rectangular/Cartesian plane consisting of two number lines intersecting at right angle at a point 'O'.
- ◆ Identify origin O and co-ordinate axes (Horizontal and Vertical axis or x-axis and y-axis respectively) in rectangular plane.
- ◆ Locate an ordered pair (a,b) as a geometrical point in the rectangular plane and recognize:
 - ◆ 'a' as the x- co-ordinate (or abscissa),
 - ◆ 'b' as the y- co-ordinate (or ordinate).
- ◆ Draw different geometrical shapes (e.g., line segment, triangle and rectangle etc.) by joining a set of given points.
- ◆ Construct a table for pairs of values satisfying a linear equation in two variables.
- ◆ Plot the pairs of points to obtain the graph of a given expression.
- ◆ Choose an appropriate scale to draw a graph.
- ◆ Draw the graph of:
 - ◆ an equation of the form $y = c$.
 - ◆ an equation of the form $x = a$.
 - ◆ an equation of the form $y = mx$.
 - ◆ an equation of the form $y = mx + c$.
- ◆ Draw a graph from the given table of values for x and y.
- ◆ Solve applied real life problems.
- ◆ Interpret conversion graph as a linear graph relating to two quantities which are in direct proportions.
- ◆ Read a given graph to know one quantity corresponding to another.
- ◆ Read the graph for conversion of the forms:
 - ◆ Miles and kilometers,
 - ◆ Acres and hectares,
 - ◆ Degrees Celsius and Fahrenheit,
 - ◆ Pakistani currency and other currencies, etc.
- ◆ Solve simultaneous linear equations in two variables by graphical method.